

Hyeonseok Jin

Ph.D. candidate at Korea Institute of Energy Technology (KENTECH)

📍 Naju, South Korea | ✉ ggyo003@kentech.ac.kr | ☎ +82-10-2502-7167

🔍 google scholar | in linkedin | 🐙 github

🔍 Research Interests

- **Computer Vision:** Generative AI, Video data analysis, Vision-Language Model
- **AI Application:** Spatio-temporal prediction, Traffic flow & accident risk forecasting, Building energy forecasting
- **Time-series Analysis:** Multi-variate forecasting, Anomaly detection

🎓 Education

Korea Institute of Energy Technology (KENTECH)

Ph.D. in Energy AI

📍 Naju, South Korea

Mar 2026 – **Present**

- GPA: -/4.3
- **Dissertation:** -
- **Advisor:** Prof. Seokju Lee

Chonnam National University

M.S. in Artificial Intelligence Convergence

📍 Gwangju, South Korea

Sep 2023 – Aug 2025

- GPA: 4.36/4.5
- **Thesis:** CNN-based Multi-In-Multi-Out Model for Efficient Spatiotemporal Prediction
- **Advisor:** Prof. Kyungbaek Kim
- **Participated projects** 📄

Chonnam National University

B.S. in Electronics and Computer Engineering

📍 Gwangju, South Korea

Mar 2016 – Feb 2023

- GPA: 3.12/4.5

🏢 Experience

Research Intern

VIEW Lab, Korea Institute of Energy Technology (KENTECH)

📍 Naju, South Korea

Sep 2025 – Feb 2026

- Research on Physics-aware Video Generation

Staff

Business Consulting & Data Team, Areum ICT Co., LTD

📍 Gwangju, South Korea

Apr 2023 – Aug 2023

- Manage the Offline Safety Zone (Data Analysis Center) in Gwangju
- Review Institutional Review Board (IRB) and Research Proposal
- Validate Data Export Codes and Files

📖 Publications (International)

*: Equal Contribution

An Improved Small Defect Classification System Using Image Patching

Jan 2026

Methodology on Aluminum Car Doors [SCIE]

IEEE Access, Vol.14, doi:10.1109/ACCESS.2026.3650927 📄

Ugur Ercelik, *Hyeonseok Jin*, Longfei Li, Kyungbaek Kim

Enhancing deep learning models for predicting smoking status using clinical data in patients with chronic obstructive pulmonary disease [SCIE]

Nov 2025

Digital Health, Vol.11, doi:10.1177/20552076251393380 [🔗](#)

Sehyun Cho, **Hyeonseok Jin**, Kyungbaek Kim, Sola Cho, Ja Yun Choi

A Data-Driven Model to Predict Regular Strength Exercise Patterns in Patients With Chronic Obstructive Pulmonary Disease: A Secondary Analysis [[SCIE](#)]

Oct 2025

IEEE Access, Vol.13, doi:10.1109/ACCESS.2025.3623702 [🔗](#)

Hyeonseok Jin, Ja Yun Choi, Sehyun Cho, Kyungbaek Kim

A Study of Leaf Disease Classification via Token-guided GNN

Aug 2025

The 14th International Conference on Smart Media & Applications (SMA 2025)

Hyeonseok Jin*, David J. Richter*, Kyungbaek Kim

Deformable Dynamic Convolution for Accurate yet Efficient Spatio-Temporal Traffic Prediction

Jul 2025

arXiv preprint, doi:10.48550/arXiv.2507.11550 [🔗](#)

Hyeonseok Jin, Geonmin Kim, Kyungbaek Kim

DATC-STP: Towards Accurate yet Efficient Spatiotemporal Prediction with Transformer-style CNN [[SCIE](#)]

May 2025

IEEE Access, Vol.13, doi:10.1109/ACCESS.2025.3573639 [🔗](#)

Hyeonseok Jin, Kyungbaek Kim

Publications (Domestic)

*: Equal Contribution

Involution-based Spatiotemporal Data Modeling for Accurate yet Scalable Traffic Accident Risk Forecasting [[Award](#)]

Jul 2025

Korea Computer Congress (KCC 2025)

Hyeonseok Jin, Kyungbaek Kim

TSPM: Two-stage Spatiotemporal Prediction Module with Floor Plan for Effective Building Energy Prediction [[Award](#)]

May 2025

Annual Symposium of KIPS (ASK 2025)

Hyeonseok Jin, Kyungbaek Kim

TCN 오토인코더 기반 VR 센서 데이터 복원 [[Award](#)]

May 2025

2025 한국전자거래학회 & 한국스마트미디어학회 춘계학술대회

진현석, 조현준, 정현수, 김경백

Enhancing Vision-Based Cigarette Smoke Detection in Smart Vehicles by Transfer Learning [[KCI](#)]

Apr 2025

Journal of Digital Contents Society, Vol.26, No.4, doi:10.9728/dcs.2025.26.4.1041 [🔗](#)

MD Ilias Bappi, **Hyeonseok Jin**, Kyungbaek Kim

Solar Power Generation Forecasting Using a Hybrid LSTM-Linear Model with Multi-Head Attention [[KCI](#)]

Feb 2025

The Transactions of the Korea Information Processing Society, Vol.14, No.2, doi:10.3745/TKIPS.2025.14.2.123 [🔗](#)

Hyeonseok Jin*, David J. Richter*, MD Ilias Bappi*, Kyungbaek Kim

건물 에너지 관리를 위한 시공간 예측 모델의 적용 가능성에 관한 연구

Oct 2024

Annual Conference of KIPS (ACK 2024)

진현석, 김경백

Web 기반 DID 메소드 동향 분석

Oct 2024

제 5회 호남 사이버보안 컨퍼런스

임도현, 김건민, 조세빈, 진현석, 박태준, 김경백

이상 전력 탐지를 위한 TCN-USAD [KCI] 스마트미디어저널, Vol.13, No.7, doi:10.30693/SMJ.2024.13.7.9  진현석, 김경백	Jul 2024
SwinVP: 비디오 예측 모델 기반 교통사고 위험 예측 2024년도 한국통신학회 하계 종합학술발표회 진현석, MD Ilias Bappi, David J. Richter, 김경백	Jun 2024
COPD 환자 운동 예측을 위한 불균형 데이터 처리 기법의 효율성에 관한 연구 Annual Symposium of KIPS (ASK 2024) 진현석, 조세현, 최자윤, 김경백	May 2024
Enhancing Speech Recognition with Whisper-tiny Model: A Scalable Keyword Spotting Approach Annual Symposium of KIPS (ASK 2024) Shivani Sanjay Kolekar, 진현석, 김경백	May 2024
이상 전력 탐지를 위한 TCN-오토인코더 [Award] 2024 한국스마트미디어학회 & 한국전자거래학회 춘계학술대회 진현석, 김경백	May 2024
분산 처리 시스템의 딥러닝 적용 가능성 검토에 관한 연구 2023년도 한국통신학회 추계 종합학술발표회 진현석, 김경백	Nov 2023
ROS 및 SLAM 기반 자율주행 로봇 시스템 2023 한국스마트미디어학회, 한국전자거래학회 스마트미디어 심포지움 진현석, 정송현, 이지훈, 김경백	Oct 2023

Honors & Awards

Best Presentation Paper Award Korea Computer Congress (KCC 2025)	Jul 2025
Excellent Paper Award Annual Symposium of KIPS (ASK 2025)	May 2025
Best Paper Award Spring Academic Conference on Smart Media 2025	May 2025
Best Paper Award Spring Academic Conference on Smart Media 2024	May 2024
Scholarship Paan Scholarship and Cultural Foundation	May 2024
3rd Place Dacon Monthly Court Judgment Prediction AI Challenge	Jul 2023

Patents & Software Registration

트랜스포머 구조의 합성곱 기반 시공간 신경망을 이용한 교통량 예측 장치 및 방법 KR Patent App, AN:10-2025-0207876 (Pending) 진현석, 김경백	Dec 2025
환자의 운동 여부 예측을 위한 인공지능 학습 및 추론 소프트웨어 KR Software Registration, GN:C-2025-053828 김경백, 진현석, 신지수	Dec 2025

하이브리드 시계열 인공지능 모델 기반 태양광 발전량 예측 웹 서비스 <i>KR Software Registration</i> , GN:C-2025-053827 김경백, David J. Richter, 진현석	Dec 2025
시공간 예측 딥러닝 모델 기반 교통 데이터 변환 및 분석 프로그램 <i>KR Software Registration</i> , GN:C-2025-053826 김경백, 진현석	Dec 2025

Academic Services

<i>IEEE Canadian Journal of Electrical and Computer Engineering</i> Journal Reviewer	2026
<i>Association for the Advancement of Artificial Intelligence (AAAI)</i> Conf. Reviewer	2026